Bidirectional Stacking Ensemble Curriculum Learning for Hyperspectral Image Imbalanced Classification With Noisy Labels

Yixin Wang, Hao Li, *Member, IEEE*, Maoguo Gong, *Fellow, IEEE*, Yue Wu, *Senior Member, IEEE*, Peiran Gong, A. K. Qin, *Fellow, IEEE*, Lining Xing, and Mingyang Zhang, *Member, IEEE*

***Abstract*— Hyperspectral imaging has demonstrated substantial advantages in enhancing classification performance in remote sensing applications due to its abundant spectral information. To address the challenges of label noise and class imbalance in the hyperspectral image (HSI) classification, we propose an end-to-end feature-guided network (FGN) for HSIs. Instead of merely combining spatial and channel attention, the FGN leverages feature-level attention interactions to enhance contextual understanding, leading to better feature extraction, especially for underrepresented classes. Furthermore, a bidirectional loss for curriculum learning (CL) is proposed to rank the HSI training data in descending or ascending order. The top and bottom loss regularizers are designed to make the proposed model suitable for noisy and imbalanced HSI data distributions. In the phase of selecting the pace parameter, a stacking ensemble CL (SECL) model is established to avoid the outliers and noisy HSI data involved in the CL training process. A novel instruction matrix based on sample weights is designed for base classifiers. The outputs of the base models, combined with the expected labels, form the input–output pairs for training the second-level classifier. Experiments conducted on multiple hyperspectral imbalanced datasets with noisy labels demonstrate the superior performance of our method.**

***Index Terms*— Curriculum learning (CL), ensemble learning, hyperspectral image (HSI) classification, imbalanced distribution.**

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Yixin Wang, Hao Li, Peiran Gong, Lining Xing, and Mingyang Zhang are with the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: yxiwang@stu.xidian.edu.cn; omegalihao@gmail.com; gpr@stu.xidian.edu.cn).

Maoguo Gong is with the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an 710071, China, and also with the Academy of Artificial Intelligence, College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010028, China (e-mail: gong@ieee.org).

Yue Wu is with the School of Computer Science and Technology, Xidian University, Xi'an 710071, China (e-mail: ywu@xidian.edu.cn).

A. K. Qin is with the Department of Computing Technologies, Swinburne University of Technology, Hawthorn, VIC 3122, Australia (e-mail: kqin@swin.edu.au).

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I. Introduction

HYPERSPECTRAL imaging involves 3-D images that contain rich spectral and spatial information [1]. Due to the abundant spectral information provided by hyperspectral images (HSIs), accurate classification can significantly enhance the understanding of complex remote sensing scenarios. HSI classification has been widely applied in various remote sensing applications, such as agricultural production [2], military reconnaissance [3], oil spill detection [4], environmental monitoring [5], and others [6].

Machine learning methods such as support vector machines [7], sparse representation [8], [9], [10], random forests [11], [12], kernel-based methods [13], [14], and deep learning [15], [16] have been pivotal in numerous HSI applications. However, their performance can deteriorate when the training data includes noisy labels. In practical remote sensing environments, the presence of noisy labels is almost unavoidable due to issues with data acquisition equipment and human annotation errors. The mislabeled data problem is further exacerbated by the complex spatial structures within large homogeneous areas. Consequently, it becomes a crucial and challenging task to develop robust models capable of handling unreliable labels in the training set.

The challenge of "noisy" labels in supervised classification has attracted increasing attention, particularly in remote sensing. Existing approaches include a unified distillation framework leveraging side information [17] and a deep self-learning framework that trains robust models without prior assumptions on label noise distribution [18]. Moreover, a spatial–spectral masked autoencoder was introduced to enhance feature learning by leveraging spatial–spectral representations [19]. Additionally, a nearest neighbor-based contrastive learning network was proposed to exploit unlabeled data, using a nearest neighbor-based augmentation strategy and a bilinear attention module to improve intermodal feature interactions [20]. Although many researchers have attempted to address the issue of noisy labels in related fields, these methods may not be directly applied to HSI classification tasks due to challenges such as imbalanced data distribution [21], [22], a lack of sufficient training samples, and high dimensionality.

In real-world remote sensing applications [23], noisy labels are unavoidable due to factors such as imprecise geolocation or errors introduced during human annotation. In particular,

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HSIs are especially susceptible to label noise due to several inherent challenges. First, HSIs often capture complex terrains, including unknown ground covers and small heterogeneous regions embedded within larger homogeneous areas, which can lead to incorrect label assignments for anomalous pixels. Second, errors in artificial labeling processes frequently stem from insufficient knowledge of ground covers or a lack of expertise in annotation. Lastly, the limited spatial resolution of hyperspectral sensors may cause a single pixel to encompass multiple ground covers, resulting in inaccurate labeling. These challenges collectively degrade the accuracy of classification results [24]. A novel clustering method based on density peak is proposed to detect noisy labels in HSI classification [25]. This method can achieve a certain level of effectiveness in identifying and rectifying noisy labels; however, it fails to account for spatial information, which is a critical factor in HSI classification tasks [26]. A novel dual-channel residual network (DCRN) is proposed that exploits intermediate spatial and spectral information to reduce the negative influence of noisy labels for HSI classification [27]. The attentive-adaptive network (AAN) [28] for HSI enhanced the effectiveness and robustness of CNNs for both clean and noisy training sets, which generate discriminative spatial–spectral features by alleviating the concentration of feature representation under the attention mechanism. However, the number of training samples per class from the above methods is the same, which is not suitable for datasets with imbalanced distribution.

Unfortunately, arising from unequal class distributions within a dataset, imbalanced distribution is a common and challenging issue in real-world HSI applications. This imbalance can negatively impact the performance of learning models, particularly when the imbalance ratio is pronounced. In such scenarios, the model may become biased toward the majority class during training, leading to a tendency to misclassify minority or rare class instances as part of the majority. This bias can have significant repercussions in HSI classification like the UP dataset, where accurately identifying instances of minority classes (e.g., Shadows) is often far more crucial than correctly classifying majority class instances (Meadows). A spectral–spatial global learning framework is introduced [29], which integrates a global convolutional long short-term memory and a global joint attention mechanism to address the challenges of insufficient and HSI imbalanced classification. Miftahushudur et al. [30] proposed utilizing the synthetic minority oversampling technique within generative adversarial networks to generate high-quality synthetic data as a solution to the imbalance issue for HSIs.

Two main challenges in HSI classification significantly impact model performance: label noise and class imbalance. Label noise arises from inherent errors in HSI annotations, making it difficult to train a robust model, especially when a significant portion of the training data contains incorrect labels. Meanwhile, the class imbalance is an unavoidable issue in real-world HSI datasets [21], [22], where certain land-cover types have significantly fewer labeled samples than others. While previous methods [27], [28] have attempted to mitigate label noise, they often assume a balanced class distribution, which is rarely the case. This limitation makes it difficult for these approaches to effectively classify categories with few samples, as the model tends to be biased toward majority classes. In this article, curriculum learning (CL) is utilized to address the above-mentioned issues. CL can effectively collect reliable samples by assigning weights to HSI training data. Furthermore, CL identifies outliers and mitigates their impact on the classifier by assigning their lower weights. Recently, CL has been proved to be effective in solving many problems, including human behavior recognition and long-term tracking [31], [32]. Self-paced CNN [33] integrated CL into convolutional architectures, achieving promising classification results in handwritten digit recognition. Hou et al. [34] applied CL in an HSI classification method for the highest confidence samples with corresponding pseudo-labels.

In this article, the FGN is proposed to generate channel-specific spatial importance maps (SIMs) in a coarse-to-fine manner, allowing the model to focus on more valuable information encoded in the features, especially the land-cover categories with few samples. Furthermore, a bidirectional loss for CL is proposed to rank the HSI training data in descending or ascending order. The bottom and top loss regularizers are designed to assign bidirectional weights to the HSI data, thereby making it suitable for noisy and imbalanced HSI data distributions. Specifically, the top loss regularizer helps filter out mislabeled or unreliable samples, preventing them from negatively impacting model learning. Meanwhile, the bottom loss regularizer ensures that minority class samples, which might be overlooked in conventional CL, receive sufficient attention during training. This design leads to improved classification performance on challenging HSI datasets. Finally, in HSI classification research, selecting the optimal pace parameter from a sequence of pace values for CL poses a significant challenge. On the one hand, increasing the pace parameter may introduce outliers and noisy HSI data into the CL training process. On the other hand, suboptimal results may be obtained if the final pace parameter is not ideal, and thus a novel SECL model with bidirectional loss is proposed to overcome the difficulties associated with pace parameter selection. The majorization minimization (MM) algorithm is employed to solve the SECL function. Experimental evaluations on multiple HSI datasets demonstrate that our newly developed method can effectively improve the performance of classifiers when fighting against noisy labels under imbalanced HSI distribution. Furthermore, it performs better on ambiguous classes (e.g., Bitumen in the UP datasets) that are often misclassified by other methods. To summarize, the main innovative contributions of the proposed approach can be summarized as follows.

1) A novel feature-guided network (FGN) for HSI classification is proposed. By leveraging the input features to guide the generation process, the FGN assigns a distinct SIM to each channel, enabling the model to focus on the most significant regions within each channel.
2) A bidirectional loss for CL is proposed to rank the HSI training data in descending or ascending order. The top and bottom loss regularizers are designed to make the proposed model suitable for noisy and imbalanced HSI data distributions.

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3) A stacking ensemble CL (SECL) model is established to address the pace parameter hard-to-be-selected issue for CL in HSI classification research. A new instruction matrix based on the sample weight is designed for classifiers, which are associated with different chosen samples.

The organization of this article is as follows. First, we briefly introduce the relevant content of CL, stacking ensemble learning in Section II. In Section III, the proposed method is described in detail. The experiments are shown in Section IV. Finally, Section V contains some conclusions and describes future work.

II. Related Work

A. Curriculum Learning

Bengio et al. [35] introduced CL, a learning paradigm inspired by human learning processes, where training starts with easy samples and progressively incorporates harder ones. The original CL formulation can be viewed as a continuation method. The key problem in CL is to define the difficultness of each sample. For example, [36] takes the negative distance to the boundary as the indicator for easiness in classification. [31] designs a new automatic CL formulation, called self-paced learning (SPL), where examples with lower losses are considered to be easier and emphasized during training.

Automatic CL dynamically adjusts the sample selection order based on the model's learning progress, continuously re-evaluating sample difficulty throughout training [37]. In [31], for instance, the likelihood of the prediction is used to rank the samples, while in [38], the objectness is considered to define the training order.

CL was successfully applied in various domains, Zhang et al. [39] used a self-paced methodology for multiple-instance learning in co-saliency detection. CL has brought improved performance levels in tasks ranging from image classification, object detection, and semantic segmentation to neural machine translation, question answering, and speech recognition. Dogan et al. [40] proposed a label similarity curriculum approach for image classification. Braun et al. [41] used anti-CL for automatic speech recognition systems under noisy environments. However, CL is not always bringing significant performance improvements. Because other factors influence performance, and these factors can be negatively impacted by CL strategies.

B. Stacking Ensemble Learning

In the last few years, ensemble models have taken more relevance due to the good performance obtained in several tasks like classification or regression problems [42], [43]. In [44], a cascade ensemble learning method is proposed and validated through experiments on complex function approximation and reliability evaluation of an aero-engine turbine rotor system. Zhang et al. [45] proposed Prompt Ensemble learning via Feedback-Reflect-Refine, a universal and automated method that overcomes the limitations of two-stage paradigms by eliminating manual prompt preparation and enabling directed optimization for weak learners. Abhishek et al. [46] presented a novel fuzzy ensemble-based deep learning algorithm using pretrained CNNs and Gompertz function optimization for five-class leukemia detection from blood smear images. These methods consist of combining different learning models to improve the results obtained by each individual model [47].

Stacking enhances model generalization by combining heterogeneous base classifiers with a second-level classifier. The general stacking method consists of three major steps. First, the first-level classifiers are created using the original training dataset. There are several strategies for learning base classifiers. For example, we can use the Bootstrap sampling technique to learn independent classifiers. To build various base classifiers, we can tune parameters in a learning algorithm. Furthermore, we can use a variety of classification and/or sampling approaches to generate base classifiers. Second, a new dataset is established from the output of the base classifiers. The labels of the first-level classifiers are considered new features in this case, but the original class labels are retained as labels in the newly generated dataset. Finally, using the newly produced dataset, create a second-level classifier. The second-level classifier could be learned using any learning approach. The main benefit of stacking ensemble is that it can shield the capabilities of a range of well-performing models to solve classification and regression problems, and it helps to prepare a better model having better predictions than all individual models.

III. Methodology

In this section, we will introduce the proposed method in detail. First, an end-to-end FGN is designed for imbalanced HSI classification tasks with noisy labels. Second, the bidirectional loss is incorporated into the CL model. Then, according to the proposed bidirectional loss, the bottom and top loss regularizers are designed to assign bidirectional weights to the HSI data. Last, an SECL algorithm is proposed to avoid selecting a proper pace parameter.

A. Feature-Guided Network

Our approach emphasizes extracting more informative features to mitigate the effects of noisy labels and imbalanced data for HSIs. To clearly illustrate the proposed FGN, the overall framework is depicted in Fig. 1 and consists of three fundamental modules.

1) Spectral Feature Derivation: To effectively manage high-dimensional spectral features, a spectral feature extraction module that performs dimensionality reduction similarly is employed for principal component analysis (PCA). Unlike PCA, this module can be trained end-to-end with other components in the network, allowing it to better adapt to the HSI classification. The module utilizes three consecutive convolutional layers as the backbone network to extract spectral features while preserving the original spatial relationships. To achieve this, each convolutional layer uses $1 \times 1 \times 7$ convolutional kernels, ensuring that the convolution operations do not alter the original spatial structure of the data. The strides for the convolutional layers are set as $(1, 1, 2)$, $(1, 1, 1)$, and $(1, 1, 1)$, respectively, with each layer employing 24 kernels.

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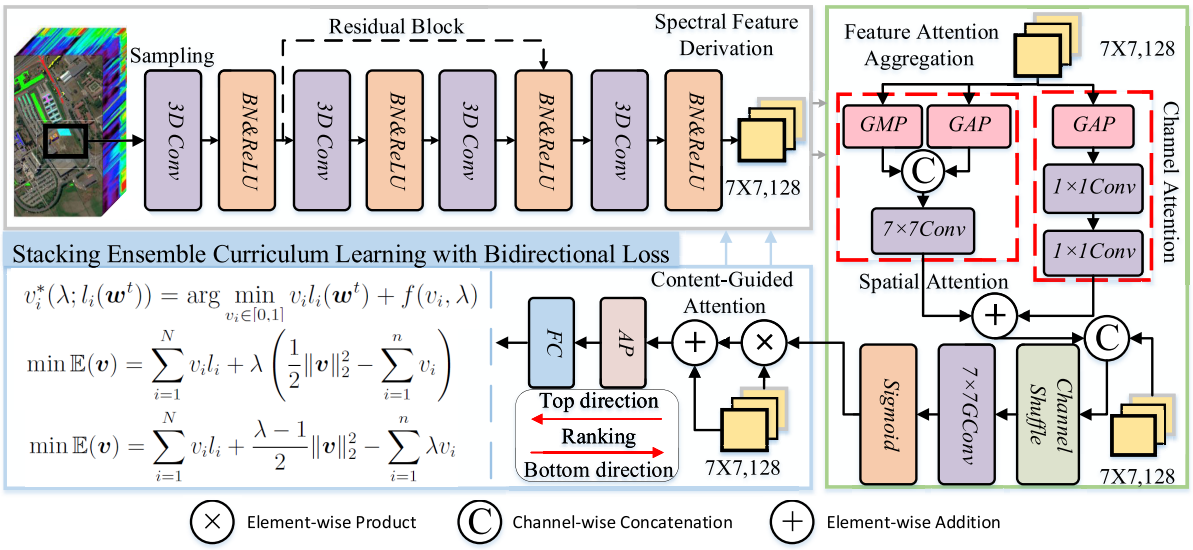


Fig. 1. Architecture of the proposed method. Initially, a spectral stem network is applied, incorporating nonadjacent residual connections between layers, facilitating robust refinement of spectral features. Subsequently, a feature attention aggregation is introduced, by assigning unique SIM to every channel, more useful information is encoded in features. Finally, a bidirectional loss function is incorporated into the CL framework, and an SECL algorithm is proposed, which optimizes the HSI learning process and enhances model robustness.

Following each convolution operation, batch normalization (BN) and the ReLU activation function are applied to stabilize the training process and enhance the feature extraction capability. To further refine the spectral features and improve robustness against noisy labels, a nonadjacent residual connection is introduced between the first and third feature maps. This residual connection provides an identity path that preserves spectral information, mitigating potential loss through convolutional transformations. After the residual block, the remaining feature maps are convolved with 128 kernels of size $1 \times 1 \times 85$ to produce 128 2-D spatial patches of size 7×7. While preserving the original spatial information, each pixel within a patch can better represent spectral information after spectral feature extraction, which improves performance against noisy labels.

2) Content-Guided Attention: The feature attention module (FAM) comprises a channel attention mechanism and a spatial attention mechanism, applied in sequence to compute attention weights along channel and spatial dimensions. The channel attention mechanism computes a channel-wise vector, $W_c \in \mathbb{R}^{C\times1\times1}$, to recalibrate features by assigning different weights to each channel. In contrast, the spatial attention mechanism generates an SIM, $W_s \in \mathbb{R}^{H\times W}$, which adaptively indicates the importance of different spatial regions.

While FAM enhances different channels and pixels selectively to improve overall performance, it has certain limitations. Specifically, the spatial attention in FAM can only capture variations at the image level, failing at the feature level. Furthermore, the channel attention in FAM focuses solely on channel-wise differences without considering contextual information. Additionally, another issue with FAM in HSI classification is the lack of interaction between the channel and spatial attention weights. The weights W_c and W_s are calculated sequentially and independently, meaning that there is no information exchange between them, leading to separate enhancement of features.

To comprehensively address these problems, a content-guided attention (CGA) mechanism that generates a unique SIM for each channel of the input features in a coarse-to-fine manner is introduced to HSI classification [48]. This approach ensures that channel attention and spatial attention weights are fully integrated, allowing for effective information interaction and better feature enhancement, which is especially necessary for unbalanced HSI data. The detailed process is depicted on the right side of Fig. 1. Let $X \in \mathbb{R}^{C\times H\times W}$ represent the input feature maps. The objective of CGA is to produce channel-specific SIMs, denoted as $W \in \mathbb{R}^{C\times H\times W}(128 \times 7 \times 7)$, which maintain the same dimensions as X. Initially, the corresponding channel attention weights W_c and spatial attention weights W_s are computed based on the methods outlined in [49] and [50]

$$\begin{aligned} W_c &= \mathcal{C}_{1\times1}\left(\max\left(0, \mathcal{C}_{1\times1}\left(X^c_{\mathrm{GAP}}\right)\right)\right) \\ W_s &= \mathcal{C}_{7\times7}\left(\left[X^s_{\mathrm{GAP}}, X^s_{\mathrm{GMP}}\right]\right) \end{aligned} \tag{1}$$

where $\max(0, x)$ represents the ReLU activation function, $\mathcal{C}_{k\times k}(\cdot)$ signifies a convolutional layer with a kernel size of $k \times k$, and $[\cdot]$ indicates the concatenation operation across channels.

X^c_{GAP}, X^s_{GAP}, and X^s_{GMP} indicate the features obtained from global average pooling over spatial dimensions, global average pooling over the channel dimension, and global max-pooling over the channel dimension, respectively. To decrease the parameter count and control model complexity, the initial 1×1 convolution reduces the channel dimension from C to (C/r) (where r is the reduction ratio), followed by a second 1×1 convolution that expands it back to C, the channel dimension is reduced to a fixed value (i.e., 16) by setting $r = (C/16)$ in this module.

Next, we combine W_c and W_s using an element-wise addition operation, which adheres to broadcasting rules, to generate

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the coarse SIMs, denoted as $W_{\text{coa}} \in \mathbb{R}^{C\times H\times W}$

$$W_{\text{coa}} = W_c + W_s. \tag{2}$$

To generate the final refined SIMs W, each channel of W_{coa} is adjusted based on the corresponding input features. The input feature content is used as guidance to produce the final channel-specific SIMs, W. Specifically, the channels of W_{coa} and X are interleaved using a channel shuffle operation as described in [51]

$$W = \sigma(\mathcal{GC}_{7\times 7}(CS([X, W_{\text{coa}}]))) \tag{3}$$

where σ denotes the sigmoid function, $CS(\cdot)$ represents the channel shuffle operation, and $\mathcal{GC}_{k\times k}(\cdot)$ refers to a group convolution layer with a kernel size of $k \times k$. In our implementation, the number of groups is set to C.

A distinct SIM is assigned to each channel, directing the model's focus toward the most relevant regions within each channel. This approach highlights critical information encoded within the features, thereby enhancing the model's performance in HSI classification tasks.

3) Subfeatures Aggregation: After being processed by CGA, 128 squares are obtained in total that are rich in feature information. Then, an average pooling layer with a 7×7 kernel is used to produce a 128-D feature vector. The average pooling layer aggregates information and reduces the overall parameter count. Then, a fully connected layer is used to generate an output prediction, $\hat{y}_i \in \mathbb{R}^C$.

B. Learning With Bidirectional Loss

In general, most HSI classification algorithms consider all the training samples uniformly, which assumes equal misclassification costs or balanced class distributions. On the one hand, it is sensitive to outliers and noise in datasets. Some recent works, such as CL [35] and SPL [31], indicate that the "easy" samples should be assigned higher costs while the "hard" ones, such as outliers or noisy samples, are given lower costs. On the other hand, imbalanced learning problems significantly compromise the performance of most learning methods. From the perspective of imbalanced learning, the "hard" samples in the minority class should be assigned large weights to protect them from being swamped by a great number of "easy" samples. To cope with the above two issues, in this article, the prior information, such as the ranking of the loss values, is considered by assigning a weight to each sample for formulating the bidirectional loss.

$\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_N, y_N)\}$ denotes the training set, where $\boldsymbol{x}_i \in \mathbb{R}^m$ is the ith training samples, m represents the number of features in $\boldsymbol{x}_i$, y_i denotes the associated label of the ith sample, and N is the number of samples in the training set. $L(y_i, g(\boldsymbol{x}_i, \boldsymbol{w}))$ represents the loss function that calculates the loss between the true label y_i and the predicted label $g(\boldsymbol{x}_i, \boldsymbol{w})$. In the decision function $g(\boldsymbol{x}_i, \boldsymbol{w})$, $\boldsymbol{w}$ represents the model parameter. In the proposed SECL model, a weight variable $\boldsymbol{v}$ is assigned to each sample. Then, we have the following bidirectional loss function:

$$\min \sum_{i=1}^{N} v_i l_i \tag{4}$$

where l_i is short for $L(y_i, g(\boldsymbol{x}_i, \boldsymbol{w}))$ and $v_i \in [0, 1]$.

To make the weight v_i have the ability to reflect the importance of the ith sample, we have the following function to describe the relationship between v_i and l_i:

$$v_i = \Phi(l_i) \tag{5}$$

where $\Phi(\cdot)$ is a monotone function. When $\Phi(\cdot)$ is a monotonically decreasing function, we have

$$\lim_{l_i\to 0} v_i = 1. \tag{6}$$

Then, (4) is called the bottom loss function since it favors "easy" samples with small losses. It means that the "easy" samples with low loss values have large weights. In general, the outliers and noisy samples always have large loss values, which are assigned lower weights to avoid polluting the learning model. When $\Phi(\cdot)$ is a monotonically increasing function, we have

$$\lim_{l_i\to 0} v_i = 0. \tag{7}$$

Then, (4) is called the top loss function since it favors "hard" samples in the minority class with large losses. It can be observed that the "easy" samples with low loss values have small weights. By giving large weights to "hard" samples in the minority class, the imbalanced learning problems can be effectively alleviated, especially for the data with separate typical and rare subpopulations.

Therefore, a bidirectional loss model with a bottom or top loss function for CL has been proposed to rank the training samples in descending or ascending order. The goal of robust CL is to reduce the influence of noisy and confusing data. The learning robustness relies on sample selection to distinguish the reliable samples from the confusing ones. When the bidirectional loss for CL ranks the training samples in descending order, the "easy" samples with low loss values have large weights, and the outliers and noisy samples always have large loss values. Obviously, it can address the noise problems. While ranking the training samples in ascending order, by giving large weights to "hard" samples in the minority class, small subsets of hard samples are protected from being swamped by large amounts of easy ones, especially for imbalanced problems. In the proposed model, each sample is assigned a weight to reflect the easiness of the sample. To further justify the effectiveness of our proposed bidirectional loss, we provide a theoretical analysis of its impact on optimization dynamics. Specifically, we examine how different weighting strategies influence sample importance during training. The detailed mathematical formulation and gradient analysis are presented in Section III-C, where we illustrate how the bottom loss suppresses the influence of noisy samples while the top loss enhances the learning of minority-class samples.

C. Bottom/Top Loss Regularizer

The relationship between the loss l_i and the weight v_i is explicitly shown in (5). To implicitly express this relationship in the loss function formula (4), we have the following optimization problem:

$$\min \mathbb{E}(\boldsymbol{w}, \boldsymbol{v}) = \sum_{i=1}^{N} v_i l_i + f(\boldsymbol{v}, \lambda) \tag{8}$$

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where λ is a regularization parameter and $f(\boldsymbol{v}, \lambda)$ is a regularization term to penalize the loss. It is obvious that the alternative search strategy can be employed to solve the above optimization problem. To better understand the impact of the Bidirectional Loss on optimization, we analyze its gradient behavior. The gradient of the loss function with respect to model parameters w is given by

$$\frac{\partial E(w, v)}{\partial w} = \sum_{i=1}^{N} v_i \frac{\partial L(y_i, g(x_i, w))}{\partial w}. \tag{9}$$

This formulation highlights the central role of v_i in controlling sample importance during training.

With the fixed $\boldsymbol{v}$, solving $\nabla_{\boldsymbol{w}} \mathbb{E}(\boldsymbol{w}) = \mathbf{0}$, the optimization problem is shown as

$$\min \mathbb{E}(\boldsymbol{w}) = \sum_{i=1}^{N} v_i l_i. \tag{10}$$

When the value of v_i is "0" or "1," it can be observed that the above optimization problem is the off-the-shelf model with a subset of the training set. When the value of v_i belongs to [0, 1], it is a weighted loss model.

To minimize the above problem with fixed model parameters $\boldsymbol{w}$, solving $\nabla_{\boldsymbol{v}} \mathbb{E}(\boldsymbol{v}) = \mathbf{0}$, we have

$$l_i + f'(v_i, \lambda) = 0. \tag{11}$$

It can be observed that (5) is equivalent to (11). Therefore, we can derive the form of $f(v_i)$ by a specified $\Phi(\cdot)$. As indicated in the above section, the function $\Phi(\cdot)$ is a monotonically decreasing or increasing function. Then, we have two types of loss regularizers that can assign proper weights to samples, benefits for addressing outliers and noise in datasets, or benefits for addressing imbalanced learning problems.

1) Bottom Loss Regularizer: To be robust to outliers and noisy samples, these samples should have little influence on the training process by giving low or zero weights to them. It is obvious that the samples with high loss values are expected to be assigned small weights. Therefore, $\Phi(\cdot)$ should be a monotonically decreasing function, the "easy" samples with low loss values have large weights, that is, the "hard" samples are given small weights. It can be uniformly realized through the following model:

$$v_i = \Phi(l_i) = 1 - \frac{\lambda}{l_i}. \tag{12}$$

The weight coefficient v_i is constrained within the range [0, 1]. When $l_i < \lambda$, the sample weight becomes zero, meaning that the corresponding samples are excluded from the training process. Conversely, for large l_i, we have $v_i \approx 0$, which ensures that noisy or highly uncertain samples contribute minimally to the gradient update. In general, the function $\Phi(l_i^*)$ is designed to be monotonically decreasing, favoring the selection of easy samples over complex ones.

As described in Section III-B, (5) equals to (11). Integrating $f'(v_i)$ in (11), we have

$$f(\boldsymbol{v}, \lambda) = \lambda \left(\frac{1}{2} \|\boldsymbol{v}\|_2^2 - \sum_{i=1}^{n} v_i \right). \tag{13}$$

2) Top Loss Regularizer: To effectively alleviate the imbalanced learning problems, "hard" samples in the minority class should be given large weights, especially for the data with separate typical and rare subpopulations. Therefore, $\Phi(\cdot)$ should be a monotonically increasing function, the "easy" samples with low loss values have small weights, that is, the "hard" samples are given large weights.

Contrary to bottom loss regularizer, $\Phi(\cdot)$ benefits for addressing the imbalanced learning problems, the function $\Phi(\cdot)$ can be easily calculated by

$$v_i = \Phi(l_i) = \frac{\lambda - l_i}{\lambda - 1}. \tag{14}$$

This ensures that important yet underrepresented samples receive stronger optimization signals, effectively balancing the learning process. The weight coefficient v_i is constrained within the range [0, 1], and according to the above formula, the sample weight becomes zero when $l_i > \lambda$. More specifically, samples with small loss values have minimal influence on the training process, allowing the learning algorithm to focus more on samples with large loss values. This mechanism is particularly beneficial for imbalanced problems, as it helps protect informative "hard" samples in the minority class from being overwhelmed by a large number of "easy" ones. In fact, in binary classification and regression tasks, top-direction loss often outperforms the average loss, which assumes an equal misclassification cost across all samples. Integrating $f'(v_i)$, we have

$$f(\boldsymbol{v}, \lambda) = \frac{\lambda - 1}{2} \|\boldsymbol{v}\|_2^2 - \sum_{i=1}^{n} \lambda v_i. \tag{15}$$

The classifier is only penalized when the predicted value related to the ground-truth class does not show up in the top most confident predicted values. Recently, hard example mining has been proposed as an effective approach in the training of deep neural networks [52] that bear resemblance to the rank-based aggregate losses. It is obvious top loss regularizer remains a convex function over all individual losses.

D. Stacking Ensemble CL

As described in Section III-C , it is hard to select the best pace parameter from the pace sequence in the current CL research. On the one hand, as the pace parameter increases, outliers and noisy samples may be incorporated into the training process of CL. On the other hand, the current CL methods may obtain bad results since the last pace parameter is not the best. Traditional stacking ensemble learning aggregates multiple base classifiers by training a second-level classifier on their outputs. However, conventional stacking methods typically treat all base classifiers' predictions equally and do not explicitly handle noisy or imbalanced data. This can lead to suboptimal performance, especially in HSI classification, where label noise and class imbalance are common challenges. Furthermore, traditional stacking does not incorporate sample selection strategies, making it sensitive to the quality of training data.

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Algorithm 1 Stacking Ensemble CL

Input: Training set $D = \{(x_1, y_1), \ldots, (x_N, y_N)\}$, Model parameters w, Regularization parameter λ, Learning rate η, Maximum iterations T, Loss function $L(y, g(x, w))$, Mode mode $\in$ {bottom, top}, Number of base classifiers n_s

Output: Optimized model parameters w, Sample weights v

Step 1: Learn first-level classifiers
Initialize weight vector $v \leftarrow 1_N$ ▷ Initialize all sample weights to 1
for $t = 1$ to n_s **do**
  Acquire the training set D_t based on the weight vector v
  Learn a base classifier h_t using D_t
end for
Step 2: Construct new dataset
Generate new dataset D' using predictions of base classifiers
Step 3: Learn second-level classifier
Train second-level classifier h_{final} on D'
Step 4: Optimize model with bidirectional loss
for $t = 1$ to T **do**
  Compute loss for each sample: $l_i \leftarrow L(y_i, g(x_i, w))$, $\forall i \in \{1, \ldots, N\}$
  Update sample weights according to mode:
  if mode == bottom **then** ▷ Bottom loss (favors easy samples)
    $v_i \leftarrow \max(0, 1 - \lambda / l_i), \forall i$
  else if mode == top **then** ▷ Top loss (favors hard samples)
    $v_i \leftarrow \max(0, (\lambda - l_i)/(\lambda - 1)), \forall i$
  end if
  Compute weighted loss: $E(w, v) \leftarrow \sum(v_i \cdot l_i) + ComputeRegularization v, \lambda$, mode
  Update model parameters: $w \leftarrow w - \eta \cdot \nabla_w E(w, v)$ ▷ Gradient descent update
  if ConvergenceCriterionMet$E(w, v)$ **then**
    break ▷ Stop if convergence criterion is met
  end if
end for
return w, v ▷ Return optimized model parameters and sample weights

Therefore, in this article, stacking ensemble learning is considered to combine multiple first-level classifiers generated by CL to avoid the selection of pace parameters. The algorithm of the proposed SECL method is shown in Algorithm 1. During iterations, different samples chosen by different pace parameters λ are as $\mathcal{D}_t$, to learn a base classifier h_t. Then, predictions of base classifiers, along with the expected outputs, provide the input and output pairs of the training dataset used to fit the second-level classifier. However, when the sample weight v is equal to 0, the corresponding sample is not involved in training. It may cause several issues when the fivefold cross-validation is considered to split the whole data. In fact, only the samples with $v > 0$ are considered to conduct the fivefold cross-validation. It can be solved by designing an instruction matrix for base classifiers.

1) Pace Sequence Based on the Sample Number: It is challenging to determine the pace parameter λ. Typically, we need to know the scope of loss values beforehand and provide the beginning value and step size with respect to the pace parameter λ to acquire a pace sequence $\{\lambda_1, \lambda_2, \ldots, \lambda_{ns}\}$, where ns is the length of the pace sequence. Instead of optimizing the proposed SECL model with respect to the pace parameter, a robust method is to explore the optimal solution with respect to the sample size used in training. Therefore, we can employ a pace sequence based on sample number $\{N_1, N_2, \ldots, N_{ns}\}$ to denote the number of selected samples in the learning process of SECL, where N_t denotes the number of samples will be selected in the tth stage. At the tth iteration, the loss l is sorted in ascending order to acquire l'. Then, the values of λ^t is defined as the N_tth loss value of l', that is, λ^t can be denoted as

$$\lambda^t = l'_{N_t} \tag{16}$$

where l' is acquired by sorting the loss l in ascending order. As described above, the selection of parameter λ in the objective function has been converted to the determination of the sample number N_t, which determines the number of samples explicitly involved in training. It is an easy task to choose several appropriate N_t to represent different λ.

2) Learn First-Level Classifiers Based on Pace Sequence: As described in Section III-D1, it is still difficult to decide the optimal $N_t (t \in [1, ns])$ for achieving the best performance of the CL-based algorithms. In this article, stacking ensemble learning is considered to address the above issue. In some stacking ensemble methods, different classifiers, such as decision trees, support vector machines, Bayes classifiers, and others, are considered the first-level classifiers. However, in this article, the FGN with different pace parameters are used as the first-level classifiers. For a pace sequence $\{\lambda_1, \lambda_2, \ldots, \lambda_{ns}\}$, ns classifiers can be constructed to form the base classifiers.

As shown in Section III-D1, when the sample weight v is equal to 0, the corresponding sample is not involved in training. It may cause several issues when the fivefold cross-validation is considered to split the whole data. On the one hand, the resulting training data may consist of too many samples with $v = 0$ and these samples are not involved in the learning process. On the other hand, since the samples with $v = 0$ are not involved in training, the corresponding predictions on these samples may not be accurate. Therefore, in this article, only the samples with $v > 0$ are considered to conduct the fivefold cross-validation. We can have the following weighted matrix $\boldsymbol{V}$:

$$\boldsymbol{V} = \begin{bmatrix} v_1^1 & v_1^2 & \cdots & v_1^{ns} \\ v_2^1 & v_2^2 & \cdots & v_2^{ns} \\ \vdots & \vdots & \ddots & \vdots \\ v_n^1 & v_n^2 & \cdots & v_n^{ns} \end{bmatrix}. \tag{17}$$

For the jth ($j \in [1, ns]$) first-level classifier, the dataset $\mathcal{D}$ with n samples is associated with a weighting vector

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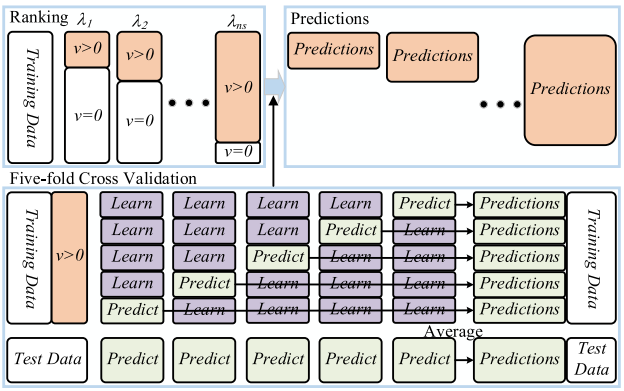


Fig. 2. Samples with the weights $v = 0$ are not involved in training first, and the base classifiers are associated with different datasets, base classifiers constituted by $\lambda_1, \lambda_2, \ldots, \lambda_{ns}$ are integrated as the first-level classifier and fivefold cross-validation is applied in this process.

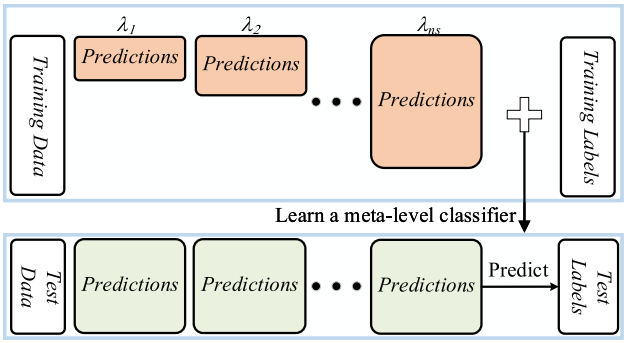


Fig. 3. Output of the first-level classifier is used as the input of the meta-level classifier, and the relationship between the input and the label y is fit. In the lower part of the picture, the output of the first stage classifier on the test set is taken as the features and passed through our learned meta-level classifier to the final predictions.

$[v_1^j, v_2^j, \ldots, v_n^j]$. Then a fivefold cross-validation method is used to split the whole data. In the fivefold cross-validation, the instances in a dataset are randomly divided into five disjoint folds of approximately equal size, and every fold is, in turn, used to test the model induced by the other fourfolds. The cross-validation process is then repeated five times.

After fivefold cross-validation, the predictions on this dataset obtained by the jth ($j \in [1, ns]$) first-level classifier are acquired. Then, we can obtain all predictions on this dataset obtained by different first-level classifiers. As described above, the samples with $v = 0$ are not involved in training. The predictions on these samples are not accurate. Therefore, we design a weighting method to calculate the average predictions on these samples, and the resulting features F_i of the ith sample for training the second-level classifier are shown as

$$F_i = \frac{\sum_{j=1}^{ns}\left(P_i^j \times \left(\boldsymbol{V}_i^j > 0\right)\right)}{\sum_{j=1}^{ns} \boldsymbol{V}_i^j > 0} \tag{18}$$

where P_i^j is the prediction on the ith sample obtained by the jth classifier. The above formula can remove the predictions on the samples with $v = 0$.

The traditional stacking models usually share a dataset. As shown in Fig. 2, in the proposed SECL model, the samples with the weights $v = 0$ are not involved in training, the base classifiers are associated with different datasets, which are the subsets of the original dataset. A weighting method is adopted to calculate the average predictions on these samples and the cross-validation is performed to ensure the data quality, which prevents the downstream prediction model from overfitting. The resulting features F_i of the ith sample and the true labels y trains the second-level classifier.

3) Learn the Second-Level Classifier: In the second phase, the output of different first-level classifiers is used to create a new feature representation for the second-level classifier. Specifically, as shown in the top of Fig. 3, a second-level classifier is learned that combines the outputs F of the first-level classifiers as inputs, with the true labels y as outputs. The output prediction of the first-level classifiers in test data is the input for the test stage of the meta-level classifiers.

F is the feature we obtained with the first-level classifier, and y is the label, thus constructing a new dataset, the linear layer is used to learn a new integrated model. After that, the output of the first stage classifier on the test set is taken as the features and passed through our meta-level classifier to the final predictions.

E. Optimization and Analysis

The proposed SECL model is solved using an MM approach. The goal of MM algorithms, which are commonly employed in machine learning, is to simplify a challenging optimization issue by alternately iterating the majorization and minimization steps. $v^*(\lambda; l)$ is the optimal solution and we can acquire the integrative function of $v^*(\lambda; l)$ as

$$F_\lambda(l) = \int_0^l \boldsymbol{v}^*(\lambda; l)dl. \tag{19}$$

$Q_\lambda(\boldsymbol{w}|\boldsymbol{w}^*)$ is used to denote a tractable surrogate for $F_\lambda(L(\boldsymbol{w}))$. Meng et al. [53] have proved that

$$\begin{aligned} Q_\lambda^{(i)}(\boldsymbol{w}|\boldsymbol{w}^*) = {} & F_\lambda(l_i(\boldsymbol{w}^*)) \\ & + v_i^*(\lambda; l_i(\boldsymbol{w}^*))(l_i(\boldsymbol{w}) - l_i(\boldsymbol{w}^*)) \end{aligned} \tag{20}$$

and we have

$$\sum_{i=1}^{n} F_\lambda(l_i(\boldsymbol{w})) \le \sum_{i=1}^{n} Q_\lambda^{(i)}(\boldsymbol{w}|\boldsymbol{w}^*). \tag{21}$$

The proof of this theorem establishes that $Q_\lambda^{(i)}(\boldsymbol{w}|\boldsymbol{w}^*)$ is a tractable substitute for $F_\lambda(l_i(\boldsymbol{w}))$ (see [53] for more details regarding this theorem). In this study, the model parameters used in the tth iteration of the MM algorithm are referred to as $\boldsymbol{w}^t$.

1) Majorization Step: $v_i^*(\lambda; l_i(\boldsymbol{w}^t))$ can be acquired by solving

$$v_i^*\left(\lambda; l_i(\boldsymbol{w}^t)\right) = \arg\min_{v_i \in [0,1]} v_i l_i(\boldsymbol{w}^t) + f(v_i, \lambda). \tag{22}$$

When $f(v_i, \lambda)$ is a bottom loss regularizer, we have

$$\min \mathbb{E}(\boldsymbol{v}) = \sum_{i=1}^{N} v_i l_i + \lambda\left(\frac{1}{2}\|\boldsymbol{v}\|_2^2 - \sum_{i=1}^{n} v_i\right). \tag{23}$$

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Then, the closed-form optimal solution is given by

$$v_i^* = \begin{cases} 1 - \dfrac{l_i}{\lambda}, & l_i < \lambda \\ 0, & l_i \geq \lambda. \end{cases} \tag{24}$$

When $f(v_i, \lambda)$ is a top loss regularizer, we have

$$\min \mathbb{E}(\boldsymbol{v}) = \sum_{i=1}^{N} v_i l_i + \frac{\lambda - 1}{2} \|\boldsymbol{v}\|_2^2 - \sum_{i=1}^{n} \lambda v_i. \tag{25}$$

Then, the closed-form optimal solution is given by

$$v_i^* = \begin{cases} 0, & l_i < \lambda \\ \dfrac{\lambda - l_i}{\lambda - 1}, & l_i \geq \lambda. \end{cases} \tag{26}$$

2) Minimization Step: In this process, we have

$$\boldsymbol{w}^{t+1} = \arg\min_{\boldsymbol{w}} \sum_{i=1}^{n} v_i L_i(\boldsymbol{w}^t). \tag{27}$$

The parameter $\boldsymbol{w}$ is learned by solving the following optimization problem:

$$\min_{\boldsymbol{w}} \frac{1}{2} \|\boldsymbol{w}\|_2^2 + C \sum_{i=1}^{n} v_i l_i \tag{28}$$

where $C(C > 0)$ represents the regularization parameter. Jiang et al. [54] showed the dual-form for solving the above problem

$$\max_{\alpha} -\frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (\phi(x_i) \cdot \phi(x_i)) + \sum_{i=1}^{n} \alpha_i \tag{29}$$

$$\text{s.t.} \sum_{i=1}^{n} \alpha_i y_i = 0 \text{ and } 0 \leq \alpha_i \leq C v_i, \quad i = 1, 2, \ldots, n \tag{30}$$

where $\phi(x_i)$ is a feature mapping function. It is obvious that a sample's upper bound is proportional to its weight.

IV. Experimental Study

In this section, the experimental settings will be introduced first. Next, we conduct comprehensive ablation studies to thoroughly evaluate the effectiveness of the proposed SECL and FGN, these experiments allow us to assess the individual contributions of each component to the overall model performance. Finally, comprehensive experiments are conducted to validate the effectiveness of our proposed method on three HSI classification tasks.

A. Description of Datasets

1) KSC [55]: The KSC dataset was collected by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over Kennedy Space Center, Florida, USA, in 1996. This dataset includes HSIs captured across a wavelength range of 400–2500 nm. The images have a spatial resolution of 512 by 614 pixels and contain 176 spectral bands after excluding those affected by water absorption and a low signal-to-noise ratio (SNR). For classification tasks, the dataset comprises 5202 labeled samples distributed across 13 land-cover classes, which reflect the diverse vegetation and surface types found in the area. The number of samples for each class in the KSC dataset is shown in Table I.

TABLE I
Classes and the Number of Samples per Class in the KSC Dataset

#	Class	Samples	#	Class	Samples
1	Scrub	761	8	Graminoid marsh	431
2	Willow swamp	243	9	Spartina marsh	520
3	Cp hammock	256	10	Cattail marsh	404
4	Slash pine	252	11	Salt marsh	419
5	Oak/Broadleaf	161	12	Mud flats	503
6	Hardwood	229	13	Water	927
7	Swamp	105			

TABLE II
Classes and the Number of Samples per Class in the UP Dataset

#	Class	Samples	#	Class	Samples
1	Asphalt	6631	6	Bare soil	5029
2	Meadows	18649	7	Bitumen	1330
3	Gravel	2099	8	Self-blocking Bricks	3682
4	Trees	3064	9	Shadows	947
5	Painted metal sheets	1345			

TABLE III
Classes and the Number of Samples per Class in the SV Dataset

#	Class	Samples	#	Class	Samples
1	Brocoli_green_weeds_1	2009	9	Soil_vinyard_develop	6203
2	Brocoli_green_weeds_2	3726	10	Corn_senesced_green_weeds	3278
3	Fallow	1976	11	Lettuce_romaine_4wk	1068
4	Fallow_rough_plow	1394	12	Lettuce_romaine_5wk	1927
5	Fallow_smooth	2678	13	Lettuce_romaine_6wk	916
6	Stubble	3959	14	Lettuce_romaine_7wk	1070
7	Celery	3579	15	Vinyard_untrained	7268
8	Grapes_untrained	11271	16	Vinyard_vertical_trellis	1807

2) UP [56]: The UP dataset was acquired using the reflective optics system imaging spectrometer (ROSIS) sensor during an airborne campaign over the University of Pavia in northern Italy. This dataset features images with a spatial resolution of 610 × 340 pixels and a ground sampling distance of 1.3 m. After discarding the noisy bands, the data comprises 103 spectral bands covering wavelengths from 430 to 860 nm. The dataset includes samples categorized into nine distinct classes, predominantly representing different stages of crop growth in the region. The number of samples for each class in the UP dataset is shown in Table II.

3) SV [57]: The SV dataset, captured over Salinas Valley, California, was obtained using the AVIRIS sensor, featuring a high spatial resolution of 3.7 m per pixel. The HSI consists of 512 rows and 217 columns. Originally containing 224 spectral bands spanning wavelengths from 400 to 2500 nm, the dataset was refined by excluding 20 bands affected by water absorption, resulting in 204 usable bands. These bands are categorized into 16 classes, which include various types of vegetation, bare soil, and vineyard fields, offering detailed insights into the agricultural landscape of the region. The number of samples for each class in the SV dataset is shown in Table III.

B. Experimental Settings

1) Parameter Setting: The proposed method is assessed using publicly available datasets, specifically KSC, UP, and SV, with artificially introduced noisy labels. These noisy labels are created by randomly substituting the true labels with incorrect ones. The training sets are made imbalanced by sampling classes in proportion to their representation within

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TABLE IV
COMPARISON OF CLASSIFICATION ACCURACY ON THE KSC DATASET: 3D-CNN, SSRN, DCRN, AAN, AND FGN

Class	260 (clean)$_{TOTAL}$ + 10 (noisy)$_{EACH}$					260 (clean)$_{TOTAL}$ + 20 (noisy)$_{EACH}$					260 (clean)$_{TOTAL}$ + 30 (noisy)$_{EACH}$					260 (clean)$_{TOTAL}$ + 40 (noisy)$_{EACH}$				
	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	FGN	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	FGN	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	FGN	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	FGN
1	81.70	82.51	**99.84**	99.12	99.52	69.69	68.61	**98.96**	98.54	98.41	64.35	54.10	**99.57**	97.48	99.09	55.57	51.22	**98.88**	93.68	98.03
2	24.31	48.31	87.79	90.01	**93.10**	16.85	33.59	79.66	80.80	**91.80**	14.13	25.20	83.70	75.64	**84.81**	12.53	21.58	68.02	66.93	**78.86**
3	28.01	45.74	**97.08**	91.02	95.99	28.30	27.13	**92.41**	88.85	87.35	21.14	23.88	71.91	85.34	**87.30**	15.46	23.23	72.57	**78.75**	66.94
4	29.73	25.48	62.93	**68.93**	68.71	14.25	21.18	51.71	**59.03**	55.88	15.12	15.25	45.88	30.57	**50.42**	12.01	17.51	**45.27**	26.83	37.08
5	13.46	31.51	50.08	55.34	**80.05**	12.66	20.76	41.30	48.28	**63.23**	9.04	19.68	**41.62**	38.03	25.72	8.42	10.26	10.09	12.93	**17.36**
6	22.78	28.72	74.93	82.52	**87.57**	15.42	20.42	84.57	68.33	**85.40**	17.34	11.16	49.68	55.83	**74.41**	12.57	14.56	26.55	31.57	**61.79**
7	24.44	25.66	**51.78**	31.44	42.54	16.48	17.11	**54.86**	16.18	18.26	12.38	11.08	10	**18.42**	13.43	8.00	13.13	**28.43**	17.88	9.21
8	37.55	59.39	98.05	**98.46**	96.71	26.33	46.45	89.74	**97.69**	95.23	23.65	35.92	**96.12**	94.86	93.49	21.56	31.31	86.88	85.55	**91.13**
9	58.80	79.43	99.19	**99.36**	98.23	50.51	65.26	**97.59**	96.46	97.39	33.14	56.99	**98.04**	96.87	96.89	30.11	47.79	**94.38**	93.43	92.78
10	36.38	58.24	99.22	**99.49**	97.01	31.45	39.68	93.55	**96.65**	95.56	22.89	30.18	93.17	**93.96**	92.61	21.11	23.47	83.96	**93.84**	92.60
11	70.73	78.00	**99.64**	97.19	99.15	55.15	53.41	**98.48**	97.25	97.27	40.59	39.44	94.88	97.60	**98.09**	35.74	34.04	88.67	**94.78**	94.23
12	54.77	56.56	**96.74**	96.01	95.31	36.03	41.33	94.72	93.02	**94.87**	32.36	35.31	**93.62**	93.01	93.57	25.45	26.64	88.80	**94.24**	93.20
13	87.71	92.94	100	100	**100**	85.61	83.80	100	100	**100**	71.54	79.01	100	100	**100**	68.63	73.77	100	100	**100**
OA	56.47	66.53	93.45	93.28	**94.53**	47.91	53.16	91.02	90.51	**91.96**	40.15	44.96	88.35	87.82	**89.72**	36.13	40.51	83.70	83.93	**86.37**
(%)	(3.85)	(3.51)	(2.01)	(3.02)	**(0.82)**	(1.33)	(2.76)	(2.06)	(2.09)	**(1.57)**	(3.58)	(2.91)	(3.45)	(1.61)	**(2.32)**	(2.37)	(4.81)	(4.90)	(1.93)	**(2.19)**
AA	43.87	54.81	85.94	85.30	**88.76**	35.29	41.44	82.89	80.08	**83.13**	29.05	33.63	75.24	75.20	**77.68**	25.17	29.88	68.66	68.49	**71.78**
(%)	(2.75)	(3.31)	(5.38)	(6.86)	**(3.16)**	(0.99)	(2.15)	(4.69)	(3.97)	**(3.40)**	(2.47)	(2.27)	(6.48)	(3.12)	**(4.91)**	(1.20)	(4.17)	(7.53)	(3.48)	**(3.47)**
KAPPA	0.51	0.62	0.92	0.92	**0.93**	0.41	0.47	0.89	0.89	**0.91**	0.33	0.38	0.86	0.86	**0.88**	0.28	0.33	0.81	0.81	**0.84**
	(0.04)	(0.03)	(0.02)	(0.03)	**(0.01)**	(0.01)	(0.03)	(0.02)	(0.02)	**(0.02)**	(0.03)	(0.03)	(0.03)	(0.02)	**(0.02)**	(2.30)	(0.05)	(0.05)	(0.02)	**(0.02)**

* M_{TOTAL} + N_{EACH}: The total number of clean samples (randomly selected) and the number of noise samples (a fixed number of noise samples per class).

the full dataset. The proposed method is implemented using PyTorch 1.13.0 and executed on a GPU with CUDA 12.2. Consistent with the approach of DCRN [27], a batch size of 16 is chosen due to the limited size of the training set. Training is managed using the ADAM optimizer with a learning rate set to 0.001. Each experiment is repeated ten times with randomly selected training samples to ensure the reliability of the results.

2) Evaluation Criteria: To thoroughly assess the performance of the proposed method, we consider three standard evaluation metrics: overall accuracy (OA) [58], average accuracy (AA) [59], and the Kappa coefficient [60]. OA refers to the proportion of correctly classified samples to the total number of samples in the dataset. A higher OA indicates better classification performance

$$OA = \frac{\sum_{i=1}^{m} h_{ii}}{N} \tag{31}$$

where N denotes the total number of samples, h_{ii} represents the number of samples correctly classified into class i, and m refers to the number of classes.

AA refers to the mean classification accuracy across all classes, providing a measure to describe the variation in classification performance for each class

$$AA = \frac{1}{m} \sum_{i=1}^{m} \frac{h_{ii}}{N_i} \tag{32}$$

where N_i represents the total number of samples for class i.

The Kappa coefficient is used to measure the consistency between the classification map and the reference image, providing a comprehensive evaluation of classification performance

$$\text{Kappa} = \frac{N \sum_{i=1}^{m} h_{ii} - \sum_{i=1}^{m} (h_{i+} h_{+i})}{N^2 - \sum_{i=1}^{m} (h_{i+} h_{+i})} \tag{33}$$

where h_{i+} represents the total number of samples in class i and h_{+i} represents the total number of samples classified as class i.

3) Compared Methods: Ten methods are used as the compared baselines. In particular, DPNLD [25], K-SDP [63], and RLPA [64] are the data-cleaning strategies, where mislabeled samples are removed to mitigate label noise. Additionally, 3-D CNN [61], SSRN [62], DCRN [27], and AAN [28] represent deep-learning models designed to build robust supervised learning frameworks capable of handling noisy labels. Furthermore, we also include three recent state-of-the-art HSI classification methods [65], [66], [67], which leverage advanced deep learning architectures to enhance classification accuracy. The relevant details are given in Supplementary Material Section I.

C. Ablation Studies

To thoroughly assess the contributions of the key components in our proposed model, we perform ablation studies on the KSC dataset. Sections IV-C1 and IV-C2 provide a detailed analysis of each component's effectiveness, supported by experimental results.

1) Test of the Noise Robustness of the FGN: The proposed FGN benefits significantly from its carefully designed network architecture, which enhances robustness against noisy labels and effectively extracts features for the categories with few samples in imbalanced HSI datasets. By utilizing input features to guide the generation of SIMs, a distinct SIM is allocated to each channel, enabling the model to focus on the most significant regions within each channel. This process highlights the more relevant information encoded in the features, thereby improving overall performance. Moreover, the inclusion of residual blocks emphasizes unique features while suppressing ambiguous ones.

The experiments were conducted using the KSC dataset, comprising 260 clean samples in all classes and varying numbers of noisy samples (10–40) in each class, to evaluate the robustness of these networks. The experimental results, summarized in Table IV, demonstrate that the FGN consistently outperforms all other methods in each experimental setting, even under high levels of label noise (up to 40 mislabeled samples per class). The ability of the FGN to maintain stable performance under such extreme conditions highlights its robustness, which is crucial for real-world applications where label noise is inevitable. While the 3-D CNN, SSRN, DCRN, and AAN all leverage spatial information for HSI

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TABLE V
COMPARISON OF CLASSIFICATION ACCURACY ON THE KSC DATASET: FGN, FGN WITH SPL, AND FGN WITH SECL

Class	260 (clean)$_{TOTAL}$ + 10 (noisy)$_{EACH}$				260 (clean)$_{TOTAL}$ + 20 (noisy)$_{EACH}$				260 (clean)$_{TOTAL}$ + 30 (noisy)$_{EACH}$				260 (clean)$_{TOTAL}$ + 40 (noisy)$_{EACH}$			
	FGN	+SPL	+SECL$_T$	+SECL$_B$	FGN	+SPL	+SECL$_T$	+SECL$_B$	FGN	+SPL	+SECL$_T$	+SECL$_B$	FGN	+SPL	+SECL$_T$	+SECL$_B$
1	**99.52**	99.22	99.49	99.41	98.41	98.39	98.96	**99.68**	99.09	**99.14**	98.34	99.05	98.03	97.10	96.42	**99.74**
2	**93.10**	91.31	90.46	86.65	**91.80**	78.56	78.97	83.78	**84.81**	76.37	71.49	81.01	**78.86**	68.76	62.21	72.05
3	**95.99**	88.48	87.83	82.59	87.35	81.61	76.33	**89.67**	**87.30**	74.71	67.64	83.30	66.94	70.89	68.91	**73.81**
4	68.71	71.65	**77.58**	76.72	55.88	61.30	**65.49**	62.24	50.42	50.02	59.01	**61.59**	37.08	42.07	50.01	**59.09**
5	80.05	68.59	70.18	**80.88**	63.23	61.16	62.03	**66.87**	25.72	36.35	46.17	**47.20**	17.36	33.50	28.00	**45.22**
6	87.57	88.51	88.15	**89.04**	**85.40**	83.85	81.54	81.95	74.41	70.80	69.26	**77.18**	61.79	64.23	60.24	**67.88**
7	42.54	66.74	74.80	**89.50**	18.26	54.41	62.56	**64.60**	13.43	35.58	33.84	**43.40**	9.21	23.36	22.41	**39.53**
8	96.71	98.35	**99.36**	97.19	95.23	**97.56**	96.73	94.74	93.49	96.42	**96.59**	93.59	91.13	**94.67**	93.53	92.82
9	98.23	96.95	99.33	**99.34**	97.39	96.77	98.56	**99.74**	96.89	93.30	**98.65**	98.25	92.78	92.21	**96.92**	96.63
10	97.01	96.07	94.64	**97.69**	**95.56**	89.05	93.21	94.61	92.61	83.67	86.76	**92.90**	92.60	75.94	82.36	**92.66**
11	**99.15**	98.33	98.77	99.00	97.27	97.80	96.77	**97.83**	98.08	97.64	95.78	**98.10**	94.23	95.91	94.47	**96.63**
12	95.31	96.56	**97.94**	96.49	**94.87**	89.01	94.01	94.26	**93.57**	88.93	92.29	92.92	**93.20**	88.29	89.48	92.63
13	100	100	100	**100**	100	100	100	**100**	100	100	100	**100**	100	99.96	100	**100**
OA	94.53	94.37	93.48	**95.24**	91.96	90.98	91.93	**93.06**	89.72	87.96	88.90	**91.09**	86.37	85.52	86.09	**89.63**
(%)	(0.82)	(1.76)	(1.41)	**(1.08)**	(1.57)	(1.29)	(1.87)	**(1.50)**	(2.32)	(1.41)	(2.46)	**(1.24)**	(2.19)	(1.63)	(1.34)	**(1.13)**
AA	88.76	89.29	88.96	**91.89**	83.13	83.81	85.01	**86.92**	77.68	77.15	78.14	**82.19**	71.78	72.84	72.69	**79.13**
(%)	(3.16)	(3.11)	(1.69)	**(2.28)**	(3.40)	(2.31)	(3.38)	**(2.84)**	(4.91)	(3.24)	(4.49)	**(3.09)**	(3.47)	(3.35)	(3.14)	**(3.21)**
KAPPA	0.93	0.93	0.93	**0.94**	0.91	0.89	0.90	**0.92**	0.88	0.86	0.87	**0.90**	0.84	0.83	0.84	**0.88**
	(0.01)	(0.01)	(0.02)	**(0.01)**	(0.02)	(0.01)	(0.02)	**(0.01)**	(0.03)	(0.01)	(0.02)	**(0.01)**	(0.02)	(0.01)	(0.01)	**(0.01)**

* M$_{TOTAL}$ + N$_{EACH}$: The total number of clean samples (randomly selected) and the number of noise samples (a fixed number of noise samples per class).

classification, their robustness to noisy labels differs significantly due to variations in network architecture. The SSRN, DCRN, and AAN incorporate residual blocks, resulting in better performance compared to the 3-D CNN. Furthermore, the FGN achieves higher accuracy than the AAN. By assigning a unique SIM to each channel, the FGN effectively focuses on the most relevant information embedded in the features. As a result, the FGN exhibits superior robustness and maintains high classification accuracy even when trained on noisy datasets with imbalanced distribution.

2) Test of the Proposed SECL Method: The SECL plays a critical role in developing a model that can better adapt to imbalanced and noisy data distributions. To systematically analyze its impact, we conduct ablation studies on the KSC dataset, using 260 clean samples in all classes while varying the number of mislabeled samples between 10 and 40 in each class. The outcomes of these experiments are summarized in Table V.

As shown in Table V, we can observe that SECL improves the quality of a reliable model in the presence of noisy labels and imbalanced data distributions. The network trained with traditional SPL tends to be more vulnerable to imbalanced data distributions. Additionally, selecting the appropriate step size introduces challenges, making the model less resilient to noisy labels easily. In contrast, the SECL framework is more adaptable and extensible to a wide range of models.

Bidirectional loss enables better adaptation to varying HSI data distributions, including imbalanced and noisy ones. Furthermore, classifiers generated by different pace parameters can be treated as weak learners. By integrating these models, the framework eliminates the need for manual pace parameter selection, further improving classification accuracy.

D. Comparison With Other Methods

The performance of the proposed method is assessed using the KSC, University of Pavia, and Salinas Valley datasets under the same network architecture and parameter configurations. For the KSC dataset, all class in the training set consists of 312 clean samples, accompanied by a varying number of mislabeled samples ranging from 4 to 12 in each class. For the Salinas dataset, all class in the training set consists of 384 clean samples, accompanied by a varying number of mislabeled samples ranging from 4 to 12 in each class, and for the University of Pavia dataset, 468 clean samples per total class and between 8 and 24 corrupted samples per class are included in the training set. Table VI presents the experimental results for the KSC dataset. The experimental results for the Salinas Valley and University of Pavia datasets are given in Supplementary Material Section II. SECL$_T$ represents the FGN with SECL$_T$ for simplicity, so is SECL$_B$. Fig. 4 illustrates the classification results, presenting the mean and standard deviation of both OA and AA across various noisy label proportions.

Traditional CNNs exhibit subpar performance across all three datasets, characterized by high variance, suggesting potential instability. In contrast, deep neural network models that include residual blocks, such as the AAN, consistently outperform the other methods, although their performance variance is elevated. Moreover, the proposed method consistently achieves higher OA and AA scores compared to the AAN and other competing methods. While the numerical improvements in OA and AA may appear modest in some cases, it is important to emphasize that these gains are achieved under noisy and imbalanced conditions where traditional methods tend to degrade significantly. Furthermore, when the proportion of noisy labels is relatively low, the advantage of SECL is less pronounced, likely due to the insufficient challenge posed by such conditions. However, as the number of noisy labels increases, SECL's superiority becomes increasingly prominent, maintaining high classification accuracy even in extreme cases.

To further illustrate the classification performance, Fig. 5 presents the classification maps generated from the Salinas Valley dataset, where only regions with ground truth are shown to enhance comparison. In general, as the proportion of noisy labels increases, model performance tends to degrade, attributable to the misleading information provided by the noisy labels, which results in incorrect predictions

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TABLE VI

CLASSIFICATION ACCURACY FOR THE KSC DATASET. CLASSIFICATION ACCURACY OBTAINED BY THE DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, AAN, AND OUR METHOD WITH 312 REAL SAMPLES AND A DIFFERENT NUMBER OF MISLABELED SAMPLES

Class	312 (clean)$_{TOTAL}$ + 4 (noisy)$_{EACH}$									312 (clean)$_{TOTAL}$ + 8 (noisy)$_{EACH}$									312 (clean)$_{TOTAL}$ + 12 (noisy)$_{EACH}$								
	DPNLD [25]	K-SDP [63]	RLPA [64]	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	SECL$_T$	SECL$_B$	DPNLD [25]	K-SDP [63]	RLPA [64]	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	SECL$_T$	SECL$_B$	DPNLD [25]	K-SDP [63]	RLPA [64]	CNN [61]	SSRN [62]	DCRN [27]	AAN [28]	SECL$_T$	SECL$_B$
1	91.45	87.94	86.33	87.39	93.74	99.43	99.71	99.39	**99.81**	78.32	85.96	84.26	81.57	90.14	99.35	99.63	98.70	**99.66**	80.75	82.89	79.01	80.07	83.23	98.24	99.48	98.23	**99.62**
2	40.18	46.12	32.11	43.40	68.13	96.31	98.10	**99.45**	95.41	35.02	36.78	29.69	32.65	59.75	91.59	95.55	**97.23**	93.53	33.12	32.05	31.97	25.80	42.42	90.74	88.45	**95.38**	93.79
3	61.90	47.72	50.48	54.11	61.93	90.43	88.33	89.52	**96.73**	41.19	41.76	43.46	36.70	44.51	83.96	**91.44**	80.63	91.23	34.46	33.64	31.33	35.88	43.67	**91.96**	85.54	88.10	89.39
4	30.40	28.33	27.56	31.91	38.55	82.84	76.68	**87.17**	86.26	24.86	24.19	26.44	32.47	28.04	77.99	75.67	81.52	**83.13**	19.91	18.67	17.29	21.08	25.28	71.32	75.61	**78.66**	76.09
5	23.07	21.99	22.04	18.55	37.25	73.45	75.36	76.24	**88.78**	19.46	18.13	14.76	15.94	27.89	68.03	80.20	63.97	**85.61**	15.09	16.33	17.08	17.00	25.08	62.61	80.10	73.19	**80.32**
6	38.28	40.59	35.36	41.10	48.60	89.22	97.23	92.75	**97.91**	29.17	40.47	30.33	31.36	34.70	93.34	87.71	94.25	**95.29**	27.04	28.22	29.00	25.88	30.02	91.55	88.87	89.87	**94.65**
7	33.40	61.90	46.42	41.83	55.35	64.32	42.78	93.19	**97.64**	23.92	37.72	30.43	28.92	44.69	72.05	30.22	73.01	**91.34**	23.23	22.24	19.52	34.20	32.80	66.44	39.01	85.67	**87.10**
8	46.66	49.04	54.02	44.50	71.36	**99.92**	99.28	98.95	98.85	40.53	41.65	44.77	41.02	62.18	**100**	97.84	98.09	98.39	32.98	34.36	32.92	40.02	65.66	**98.80**	98.29	97.06	98.37
9	75.27	83.41	68.93	77.51	91.07	98.30	99.85	99.52	**99.97**	65.92	72.34	63.48	72.07	85.80	97.94	98.30	97.03	**99.28**	58.08	61.26	64.02	61.56	83.51	98.17	94.30	99.10	**100**
10	64.83	59.45	65.21	64.58	88.01	99.92	98.46	**99.94**	99.51	48.32	50.08	45.20	53.75	78.59	**99.49**	99.25	98.69	99.23	41.65	47.93	40.17	49.95	68.06	**99.61**	97.92	96.44	98.46
11	84.46	85.53	79.69	85.18	87.84	99.59	99.46	99.17	**99.84**	76.23	77.66	73.28	72.42	78.87	99.50	99.40	99.55	**99.73**	75.60	70.72	66.36	65.34	72.62	99.37	98.88	97.36	**99.78**
12	76.09	69.86	71.35	72.08	75.04	97.09	97.19	98.78	**99.54**	56.78	59.35	60.89	62.62	72.05	96.69	95.37	**99.15**	99.02	48.82	52.21	51.03	52.12	51.63	95.16	95.19	98.03	**99.55**
13	98.05	97.19	92.90	95.87	88.91	100	100	100	**100**	91.50	94.59	89.32	94.97	95.28	100	100	**100**	99.95	92.33	90.43	87.34	80.06	92.04	100	100	**100**	100
OA	70.92	70.44	67.30	69.61	78.20	95.97	95.69	97.27	**98.27**	60.50	64.19	60.95	63.03	73.08	95.32	94.83	95.36	**97.24**	57.54	58.22	56.03	56.22	66.88	94.70	93.96	95.46	**96.66**
(%)	(1.51)	(2.97)	(2.98)	(3.02)	(6.68)	(2.03)	(1.98)	(1.33)	**(0.61)**	(3.43)	(1.78)	(2.69)	(2.89)	(4.27)	(1.70)	(1.55)	(3.12)	**(0.94)**	(3.12)	(2.32)	(2.42)	(3.78)	(3.39)	(2.42)	(1.67)	(0.89)	**(1.41)**
AA	58.77	59.93	56.28	58.31	69.68	91.60	90.19	94.93	**96.94**	48.56	52.36	48.95	50.50	61.73	90.76	88.51	90.91	**95.03**	44.85	45.46	43.62	45.31	55.08	89.54	87.82	92.08	**93.62**
(%)	(2.52)	(3.95)	(3.52)	(4.09)	(4.58)	(4.51)	(5.11)	(2.77)	**(1.12)**	(2.53)	(2.59)	(2.42)	(2.57)	(4.29)	(3.83)	(3.26)	(6.20)	**(1.89)**	(3.15)	(2.03)	(2.78)	(2.87)	(2.82)	(5.02)	(1.74)	(2.02)	**(2.34)**
KAPPA	0.67	0.66	0.63	0.66	0.75	0.95	0.95	0.96	**0.98**	0.55	0.59	0.56	0.58	0.69	0.94	0.94	0.94	**0.96**	0.52	0.53	0.50	0.51	0.63	0.94	0.93	0.94	**0.96**
	(0.02)	(0.03)	(0.03)	(0.03)	(0.07)	(0.02)	(0.02)	(0.01)	**(0.01)**	(0.04)	(0.02)	(0.04)	(0.03)	(0.04)	(0.02)	(0.02)	(0.03)	**(0.01)**	(0.03)	(0.02)	(0.03)	(0.04)	(0.03)	(0.03)	(0.02)	(0.01)	**(0.01)**

* M$_{TOTAL}$ + N$_{EACH}$: The total number of clean samples (randomly selected) and the number of noise samples (a fixed number of noise samples per class).

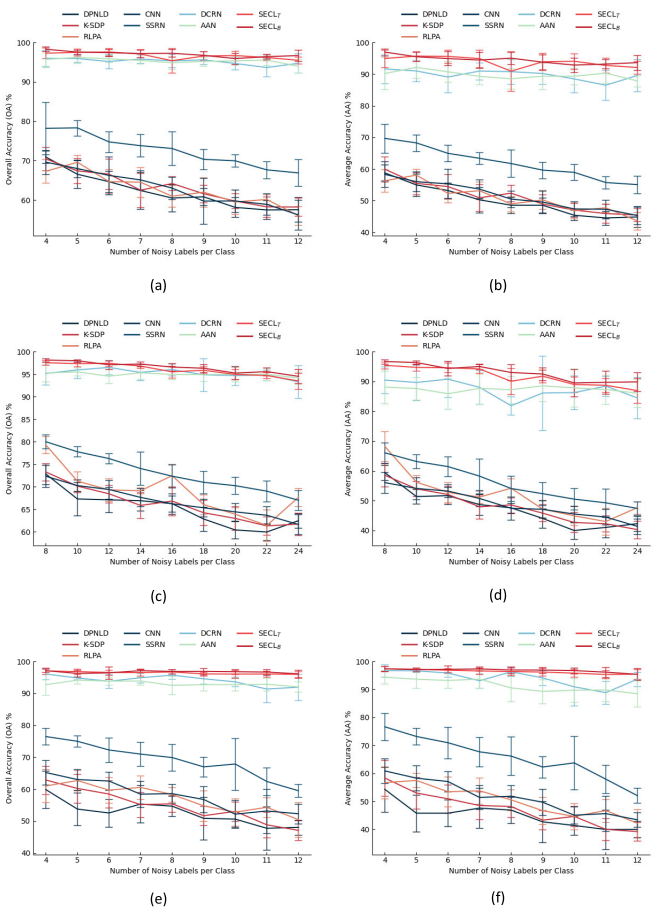


Fig. 4. Performance comparison of several methods, including the DPNLD, K-SDP, RLPA, 3-D CNN, SSRN, DCRN, AAN, SECL$_T$, and SECL$_B$, in terms of OA (first row) and AA (second row). (a) and (b) Trained with the KSC dataset with 312 clean samples in all classes and various mislabeled samples in each class from 4 to 12. (c) and (d) Trained with the UP dataset with 468 clean samples in all classes and various mislabeled samples in each class from 8 to 24. (e) and (f) Trained with the Salinas Valley dataset with 384 clean samples in all classes and various mislabeled samples in each class from 4 to 12.

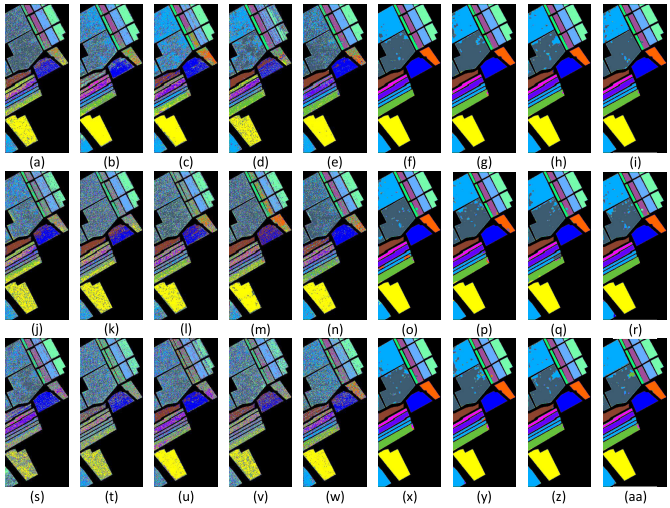


Fig. 5. Classification map for the Salinas Valley dataset. Classification map obtained by DPNLD (first column), K-SDP (second column), RLPA with extreme learning machine (ELM) classifier (third column), 3-D CNN (fourth column), SSRN (fifth column), DCRN (sixth column), AAN (seventh column), SECL$_T$ (eighth column), and SECL$_B$ (last column). (a)–(i) Four mislabeled samples for each class. (j)–(r) Eight mislabeled samples for each class. (s)–(aa) Twelve mislabeled samples for each class.

and reduced generalization ability. As depicted in Fig. 5, our method exhibits clearer boundaries and more uniform regions compared to other methods, further highlighting its advantages. Experiments were also conducted to assess the performance when the training set contained fewer clean samples. The relevant details are given in Supplementary Material Section III.

Compared to our method, the performance of the 3-D CNN, SSRN, DCRN, and AAN is notably suboptimal. The poor performance of these models can be attributed to the complex structure of the training sets, where a small number of clean samples combined with an imbalanced data distribution hampers the ability of models to correctly interpret the training signals. Our method demonstrates superior performance, particularly under conditions of limited training data, more effective for HSI classification.

E. Comparison With the Latest SOTA Methods

To further validate the effectiveness of our proposed SECL method, we conducted additional experiments comparing it with the latest state-of-the-art methods for HSI classification, including the CSCANet, DBCTNet, MASSFormer, and AAN. Among these methods, the AAN is specifically designed for noisy HSI classification, whereas the others primarily address general HSI classification. The results, provided in

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the Supplementary Material Section IV, show that SECL achieves superior performance across the board, highlighting its robustness. While the newly added methods demonstrated competitive results, their performance on the tested datasets did not surpass that of the AAN, further reinforcing the effectiveness of our approach.

V. Concluding Remarks

In this article, a novel approach is proposed to address label noise and imbalanced distribution in HSI classification. The proposed FGN for HSIs integrates feature-level attention interactions, while SECL effectively handles imbalanced HSI data distributions and varying proportions of noisy labels. The proposed method is evaluated on three benchmark datasets, consistently outperforming state-of-the-art approaches. Despite these promising results, we acknowledge that the ensemble strategy in SECL introduces additional computational overhead compared to single-model approaches. Future work could explore more efficient strategies to reduce computational costs while maintaining classification performance. Moreover, future research may explore the integration of additional types of remote sensing data, such as light detection and ranging (LiDAR) and synthetic aperture radar (SAR), with hyperspectral data to enhance the adaptability and noise resilience of classification models in complex terrain scenarios. The fusion of multisource data is expected to enhance feature representations, contributing to improved classification accuracy in challenging scenarios.

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Yixin Wang received the B.Eng. degree from Xidian University, Xi'an, China, in 2021, where he is currently pursuing the Ph.D. degree in electronic science and technology with the School of Electronic Engineering.

His research interests include machine learning, computer vision, and intelligent interpretation of remote sensing images.

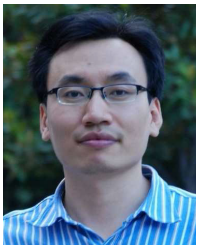


Hao Li (Member, IEEE) received the B.Eng. degree in electronic engineering and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2013 and 2018, respectively.

He is currently an Associate Professor with the School of Electronic Engineering, Xidian University. His research interests include computational intelligence and machine learning.

Dr. Li was a member of the IEEE CIS Neural Networks Technical Committee. He was a recipient of the Young Distinguished Talent, Shaanxi, China, and the Young Talent of the University Association for Science and Technology, Shaanxi. He received the Excellent Doctoral Dissertation Award of Shaanxi Province in 2021.

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Maoguo Gong (Fellow, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2003 and 2009, respectively.

Since 2006, he has been a Teacher at Xidian University. He was promoted to an Associate Professor and a Full Professor in 2008 and 2010, respectively, both with exceptive admission. He has published over 100 articles in journals and conferences and holds over 20 granted patents as the first inventor. He is leading or has completed over 20 projects as the Principal Investigator, funded by the National Natural Science Foundation of China, the National Key Research and Development Program of China, and others. His research interests include computational intelligence, with applications to optimization, learning, data mining, and image understanding.

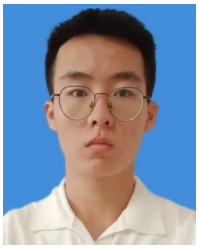
Dr. Gong was a recipient of the prestigious National Program for Support of the Leading Innovative Talents from the Central Organization Department of China, the Leading Innovative Talent in Science and Technology from the Ministry of Science and Technology of China, the Excellent Young Scientist Foundation from the National Natural Science Foundation of China, the New Century Excellent Talent from the Ministry of Education of China, and the National Natural Science Award of China. He is the Executive Committee Member of the Chinese Association for Artificial Intelligence, a Senior Member of the Chinese Computer Federation, and an Associate Editor or Editorial Board Member for over five journals including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



Yue Wu (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2016, respectively.

Since 2016, he has been a Teacher with Xidian University, where he is currently a Professor. He has authored or co-authored more than 90 articles in refereed journals and proceedings. His research interests include computational intelligence and its applications.

Dr. Wu is the Secretary General of the Chinese Association for Artificial Intelligence-Youth Branch, Chair of CCF YOCSEF Xi'an, and a Senior Member of the Chinese Computer Federation. He is an Editorial Board Member for over five journals, including *Remote Sensing*, *Applied Sciences*, *Electronics*, and *Mathematics*.



Peiran Gong received the B.Eng. degree from Nanjing Tech University, Nanjing, China, in 2021. He is currently pursuing the Ph.D. degree from the School of Electronic Engineering, Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, Xidian University, Xi'an, China.

His research interests include deep learning and computer vision.



A. K. Qin (Fellow, IEEE) received the B.Eng. degree from Southeast University, Nanjing, China, in 2001, and the Ph.D. degree from Nanyang Technological University, Singapore, in 2007.

From 2007 to 2017, he was with the University of Waterloo, Waterloo, ON, Canada, INRIA Grenoble Rhône-Alpes, Montbonnot-Saint-Martin, France, and RMIT University, Melbourne, VIC, Australia. In 2017, he joined Swinburne University of Technology, Hawthorn, VIC, Australia, where he is now a Professor. He currently serves as the Director of the Swinburne Intelligent Data Analytics Laboratory and the Deputy Director of the Swinburne Space Technology and Industry Institute. Also, he is the Vice President for Education at the IEEE Computational Intelligence Society (CIS). His major research interests include machine learning, evolutionary computation, computer vision, remote sensing, services computing, and edge computing.

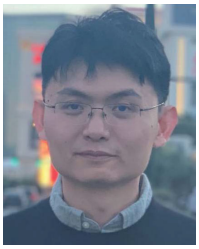
Dr. Qin was a recipient of the 2012 IEEE Transactions on Evolutionary Computation Outstanding Paper Award and the 2022 IEEE Transactions on Neural Networks and Learning Systems Outstanding Associate Editor. He is currently the Chair of the IEEE CIS Neural Networks Task Force on "Deep Vision in Space," the Vice-Chair of the IEEE CIS Emergent Technologies Task Force on "Multitask Learning and Multitask Optimization," and the Vice-Chair of the IEEE CIS Neural Networks Task Force on "Deep Edge Intelligence." He served as the General Co-Chair of the 2022 IEEE International Joint Conference on Neural Networks (IJCNN) and as the Chair of the IEEE CIS Neural Networks Technical Committee from 2021 to 2022.



Lining Xing received the double B.S. degree in economics and science from Xi'an Jiaotong University, Xi'an, China, in 2002, and the Ph.D. degree in management science from the National University of Defense Technology, Changsha, China, in 2009.

He is currently a Professor with the School of Electronic Engineering, Xidian University, Xi'an. His research interests include evolutionary algorithms and combinatorial optimization.

Dr. Xing was a recipient of the National Excellent Ph.D. Dissertation of China Award, in 2014, the Excellent Talent Support Program in New Century from the Ministry of Education, China, and the Funds for Distinguished Young Scientists from the Natural Science Foundation of Hunan.



Mingyang Zhang (Member, IEEE) received the B.Eng. degree in automation and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2012 and 2018, respectively.

Since 2018, he has been a Lecturer with the School of Electronic Engineering, Xidian University. His research interests include computational intelligence and remote sensing image understanding.

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